MATH 8090: Time Series Regression: Unit Roots, Spurious Regression, and Prewhitening

Readings: CC08 Chapters 3.3–3.4, 6.4, and 11.3–11.4; BD16 Chapters 6.3, 6.6, and 11.1; SS17 Chapters 2.1, 3.8, 5.2, and 5.5

Whitney Huang



School of
MATHEMATICAL AND
STATISTICAL SCIENCES
Clemson University

Week 9

Agenda

Time Series Regression Models

Generalized Least Squares Regression

Unit Root Tests in Time Series Analysis

Spurious Correlation and Prewhitening

Why Time Dependence Changes Standard Statistical Methods

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regression, tests, correlation.

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Regression	$\Longrightarrow$	Correlated errors / GLS
Stationarity	$\Longrightarrow$	Unit-root tests
Correlation between series	$\Longrightarrow$	Prewhitening

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Stationarity	$\Longrightarrow$	Unit-root tests
Correlation between series	$\Longrightarrow$	Prewhitening

Central question: What goes wrong if we ignore temporal dependence?

Regression with Time-Series Errors

Basic Model

$$Y_t = \underbrace{m_t}_{\text{systematic mean}} + \underbrace{\eta_t}_{\text{stationary error}}$$

Assume $E(\eta_t) = 0$ and $\{\eta_t\}$ is weakly stationary:

$$E(Y_t) = m_t, \qquad \mathrm{Cov}(Y_t, Y_{t+h}) = \gamma_\eta(h).$$

- **Regression:** models the systematic mean m_t.
- **Time-series model:** captures temporal dependence remaining in η_t.

Key Idea

Regression explains the mean; the time-series model explains the remaining temporal dependence.

Modeling the Mean

A common specification is

$$m_t = \beta_0 + \beta_1 x_{1t} + \cdots + \beta_p x_{pt}.$$

Examples:

- Constant mean: $m_t = \beta_0$
- Linear trend: $m_t = \beta_0 + \beta_1 t$
- Harmonic regression:

$$m_t = \beta_0 + \sum_{k=1}^{K} \left[a_k \cos\left(\frac{2\pi k t}{s}\right) + b_k \sin\left(\frac{2\pi k t}{s}\right) \right]$$

Matrix notation:

$$\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \qquad \mathrm{Var}(\boldsymbol{\eta}) = \boldsymbol{\Sigma}}$$

What Does OLS Assume About the Errors?

Consider

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \qquad E(\boldsymbol{\eta} \mid \mathbf{X}) = \mathbf{0}.$$

Ordinary least squares:

$$\widehat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Classical Error Assumption

$$\mathrm{Var}(\boldsymbol{\eta} \mid \mathbf{X}) = \sigma^2 \mathbf{I}$$

Errors are uncorrelated with constant variance.

With time-series errors:

$$\mathrm{Var}(\boldsymbol{\eta} \mid \mathbf{X}) = \boldsymbol{\Sigma} \neq \sigma^2 \mathbf{I}.$$

What happens if we ignore the off-diagonal covariances?

What Happens If We Ignore Autocorrelation?

Suppose

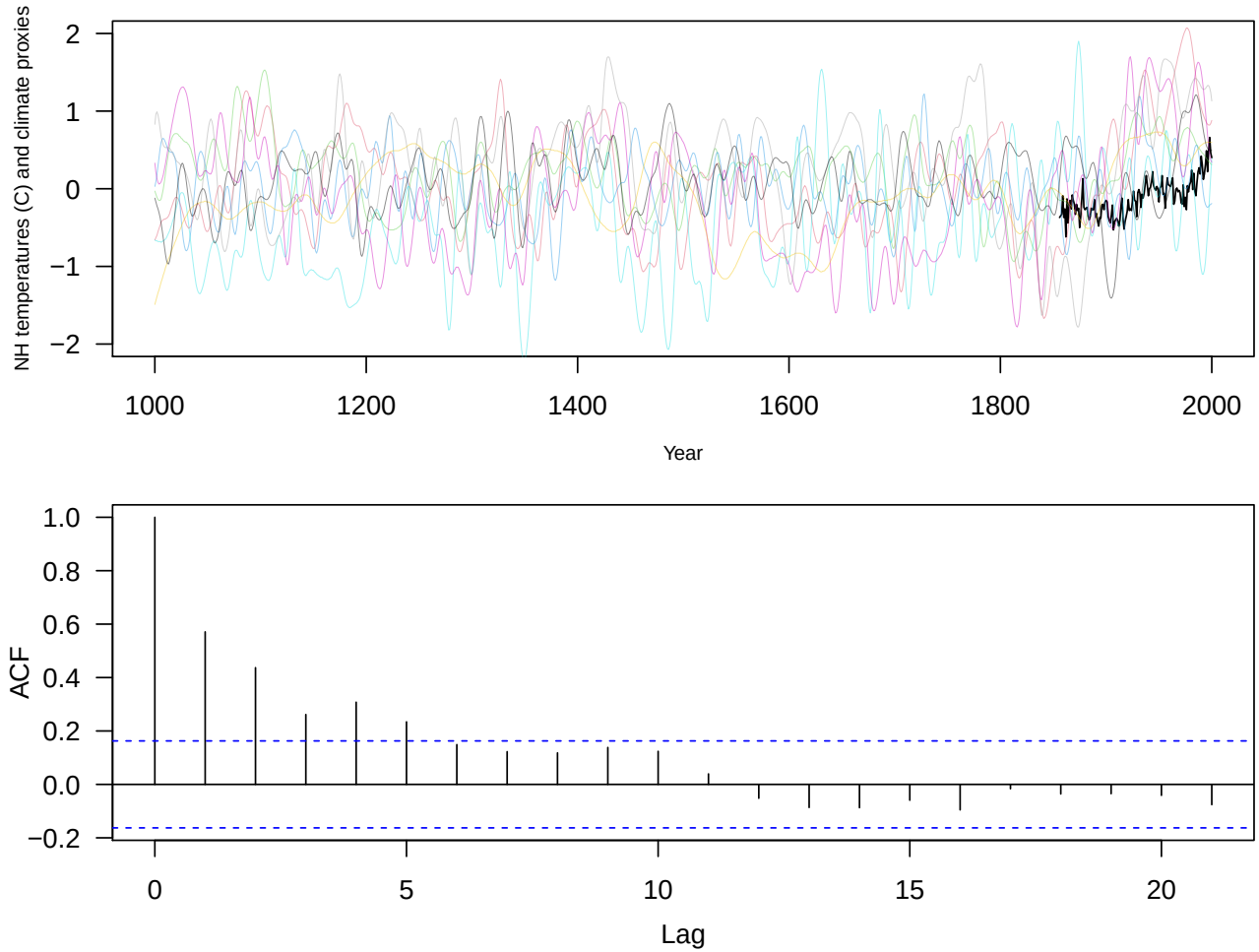
$$E(\boldsymbol{\eta} \mid \mathbf{X}) = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\eta} \mid \mathbf{X}) = \boldsymbol{\Sigma}.$$

- **Point estimates:** OLS remains conditionally unbiased under exogeneity.
- **Efficiency:** OLS is generally no longer the best linear unbiased estimator.
- **Inference:** Conventional OLS standard errors, confidence intervals, and tests may be incorrect.

Correct OLS Covariance

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}}_{OLS} \mid \mathbf{X}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Temperatures and Tree Ring Proxies [Jones & Mann, 2004]



Key Observation

Residual autocorrelation suggests that conventional i.i.d.-error OLS inference may be unreliable.

Why Generalized Least Squares?

Consider

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \qquad \mathrm{Var}(\boldsymbol{\eta}) = \boldsymbol{\Sigma}.$$

OLS vs. GLS

$$\text{OLS:} \quad \min_{\beta} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^{\top} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

$$\text{GLS:} \quad \min_{\beta} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^{\top} \boldsymbol{\Sigma}^{-1} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

- OLS treats residual directions equally.
- GLS accounts for error covariance through $\boldsymbol{\Sigma}^{-1}$.

Key Idea

GLS transforms correlated errors into uncorrelated errors before regression.

GLS Estimation and Inference

Assume

$$\boldsymbol{\eta} \sim N(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} > 0.$$

GLS Estimator

$$\widehat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{Y}.$$

Sampling distribution:

$$\widehat{\boldsymbol{\beta}}_{GLS} \sim N\left(\boldsymbol{\beta}, (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1}\right).$$

- GLS is BLUE when $\boldsymbol{\Sigma}$ is known.
- GLS reduces to OLS if $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$.
- In practice, estimate $\boldsymbol{\Sigma}$ using an ARMA model (feasible GLS).

Applying GLS in Practice

Challenge: The error covariance matrix $\boldsymbol{\Sigma}$ is unknown and must be estimated.

Two-Step Approach: Feasible GLS

1. Fit OLS and obtain residuals:

$$\widehat{\boldsymbol{\eta}} = \mathbf{Y} - \mathbf{X}\widehat{\boldsymbol{\beta}}_{OLS}.$$

2. Fit an ARMA model to the residuals to estimate $\widehat{\boldsymbol{\Sigma}}$, then refit using GLS:

$$\widehat{\boldsymbol{\beta}}_{FGLS} = (\mathbf{X}^\top \widehat{\boldsymbol{\Sigma}}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \widehat{\boldsymbol{\Sigma}}^{-1} \mathbf{Y}.$$

One-Step Approach: Maximum Likelihood

Jointly estimate regression coefficients and ARMA error parameters.

Likelihood-Based Regression Methods

Model:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \sim \mathrm{N}(\boldsymbol{0}, \Sigma)$

$$\Rightarrow \boldsymbol{Y} \sim \mathrm{N}(\boldsymbol{X}\boldsymbol{\beta}, \Sigma)$$

We maximize the Gaussian likelihood

$$\begin{aligned} & L_n(\boldsymbol{\beta}, \boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) \\ & = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2} (\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta})^T \Sigma^{-1} (\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}) \right] \end{aligned}$$

with respect to the regression parameters $\boldsymbol{\beta}$ and ARMA parameters $\boldsymbol{\phi}$, $\boldsymbol{\theta}$, σ^2 simultaneously

Two-Step vs. One-Step Estimation

Two-Step: GLS	**One-Step: MLE**
Fit mean model by OLS	Fit mean and dependence jointly
Estimate ARMA from residuals	Estimate all parameters simultaneously
Refit regression using GLS	Maximize Gaussian likelihood
Simple and interpretable	Unified likelihood-based approach

Two-step workflow:

$$\boxed{\text{OLS} \rightarrow \widehat{\eta}_t \rightarrow \text{ARMA} \rightarrow \widehat{\Sigma} \rightarrow \text{GLS}}$$

One-step workflow:

$$\boxed{\boldsymbol{Y} \xrightarrow{\text{Joint MLE}} (\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\phi}}, \widehat{\boldsymbol{\theta}}, \widehat{\sigma}^2)}$$

Monte Carlo Comparison: Two-Step vs. One-Step

Data-generating model:

$$Y_t = 3 + 0.5x_t + \eta_t,$$

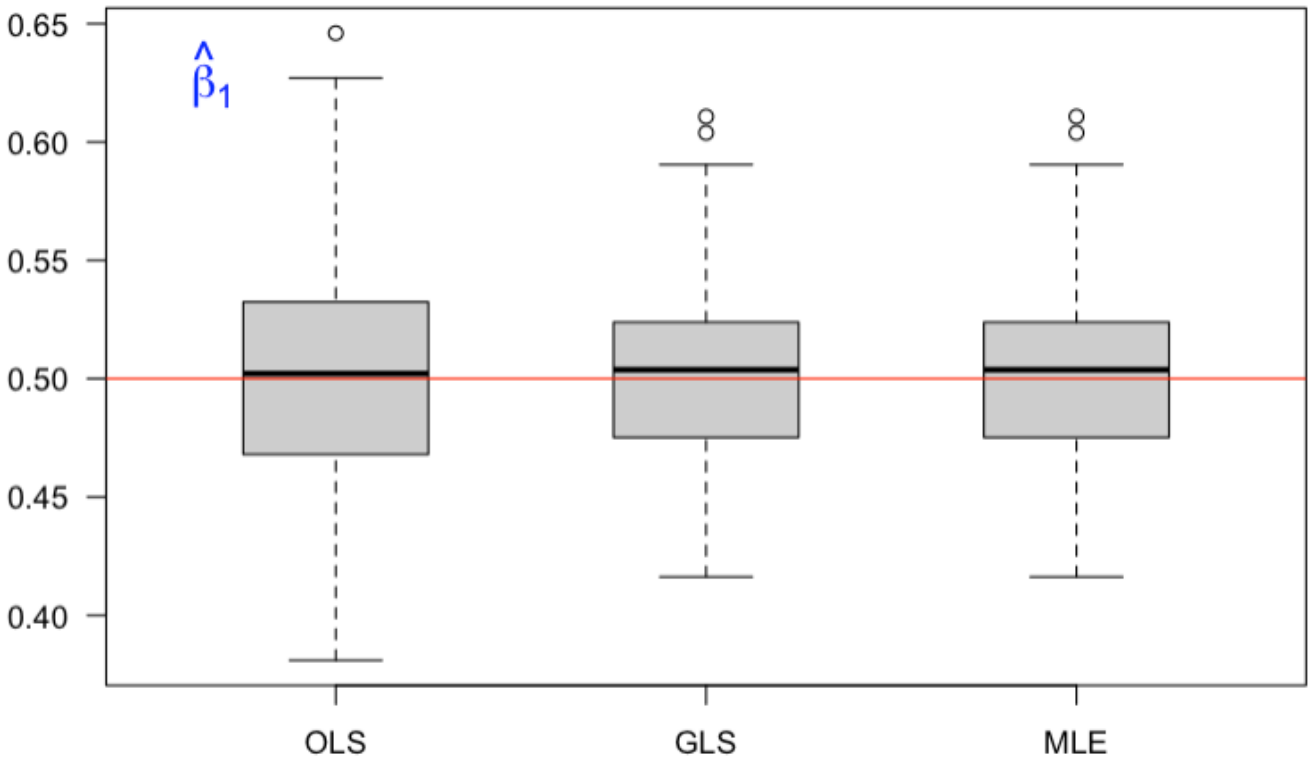
where $\{\eta_t\}$ follows an ARMA(1,1) process:

$$\eta_t = 0.8\eta_{t-1} + Z_t - 0.4Z_{t-1}, \qquad Z_t \overset{iid}{\sim} N(0,1).$$

Simulation design:

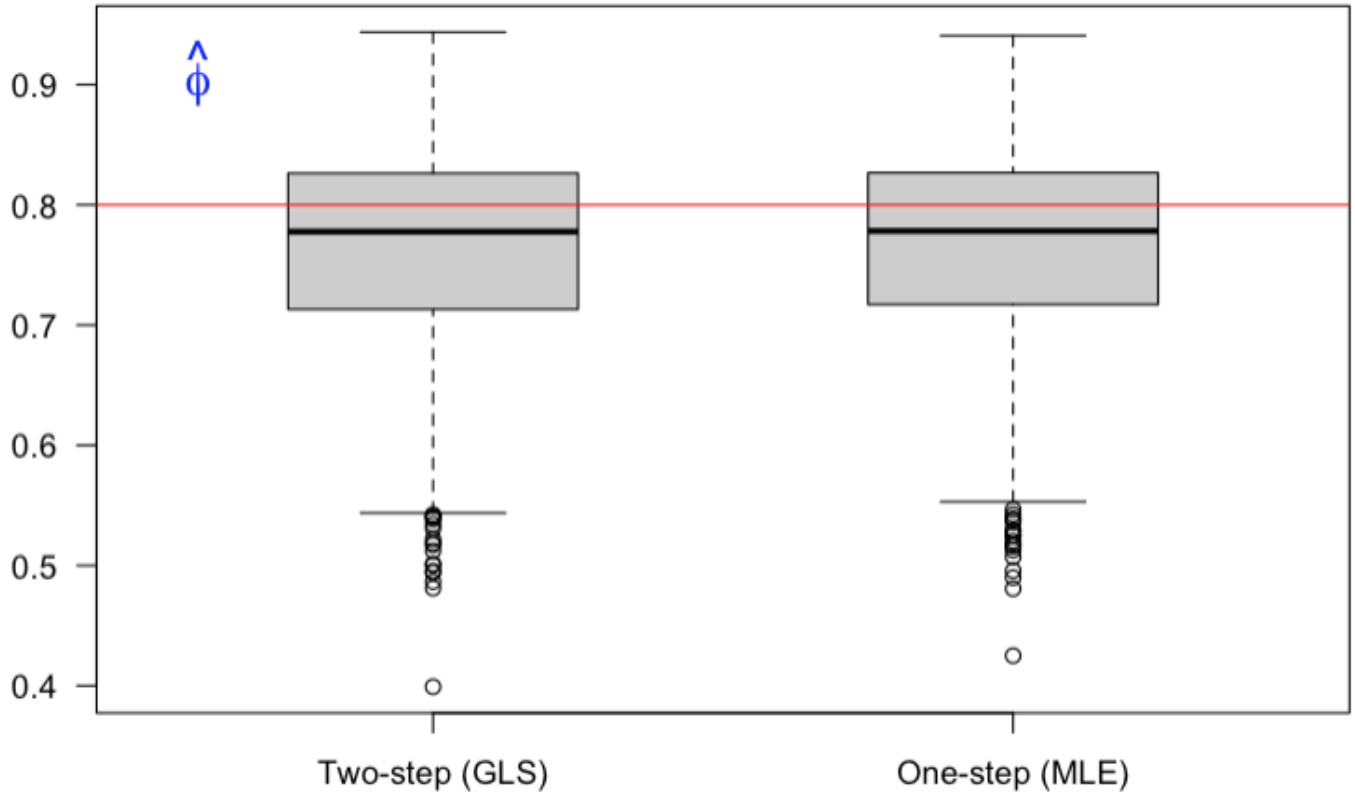
1. Generate 500 replications, each of length $n = 200$.
2. **Two-step:** Fit OLS, estimate ARMA(1,1) from residuals, then refit using GLS.
3. **One-step:** Jointly estimate regression and ARMA parameters by MLE.
4. Compare bias, standard deviation, confidence interval coverage, and width.

Comparing Regression Slope Estimates



Method	OLS	GLS	MLE
Bias	-4e-4	9e-4	9e-4
Sd	0.046	0.035	0.035
CI coverage	90.8%	93.6%	93.6%
CI width	0.162	0.129	0.129

Comparing ARMA Estimates



Method	GLS	MLE
Bias	-0.038	-0.036
Sd	0.090	0.089
CI coverage	96.6%	96.2%
CI width	0.330	0.328

Simulation Study: What Did We Learn?

- All three methods yield nearly unbiased estimates of the regression slope.
- Ignoring serial correlation leads to **greater sampling variability** and **poorer confidence interval coverage**:

$$\begin{aligned} \text{SD:} \quad & 0.046\ (\text{OLS}) \ \rightarrow\ 0.035\ (\text{GLS}), \\ \text{Coverage:} \quad & 90.8\%\ (\text{OLS}) \ \rightarrow\ 93.6\%\ (\text{GLS}). \end{aligned}$$

- Two-step GLS and one-step MLE perform almost identically in this simulation.

Key Takeaway

Modeling temporal dependence matters more here than the choice between two-step GLS and one-step MLE.

Case Study: Lake Huron Water Levels

Data: Annual Lake Huron water levels, 1875–1972.

Regression with Time-Series Errors

$$Y_t = \beta_0 + \beta_1 t + \eta_t, \qquad \{\eta_t\} \sim \text{stationary ARMA}(p, q).$$

Scientific question:
Has the mean water level changed linearly over time?

Statistical hypothesis:

$$H_0 : \beta_1 = 0 \qquad \text{vs.} \qquad H_1 : \beta_1 \neq 0.$$

Key Challenge

Account for temporal dependence when estimating the trend and assessing its significance.

Lake Huron: Two-Step Estimation

Step 1: Fit OLS

```
lm(formula = LakeHuron ~ years)

Residuals:
     Min       1Q   Median       3Q      Max
-2.50997 -0.72726  0.00083  0.74402  2.53565

Coefficients:
             Estimate Std. Error t value
(Intercept) 625.554918   7.764293  80.568
years        -0.024201   0.004036  -5.996
```

Step 2: Fit ARMA to Residuals

```
arima(x = lm$residuals, order = c(2, 0, 0), include.mean = FALSE)

Coefficients:
         ar1      ar2
      1.0050  -0.2925
s.e.  0.0976   0.1002
```

Step 3: Refit Using GLS

Use the estimated error covariance $\widehat{\Sigma}$ to re-estimate the trend.

Exercise: Complete the GLS refitting step.

Lake Huron: One-Step Maximum Likelihood

Jointly estimate the regression trend and ARMA error parameters.

```
> mle <- arima(LakeHuron, order = c(2, 0, 0),
+              xreg = cbind(rep(1,length(LakeHuron)), years),
+              include.mean = FALSE)
> mle

Call:
arima(x = LakeHuron, order = c(2, 0, 0), xreg = cbind(rep(1, length(LakeHuron)),
    years), include.mean = FALSE)

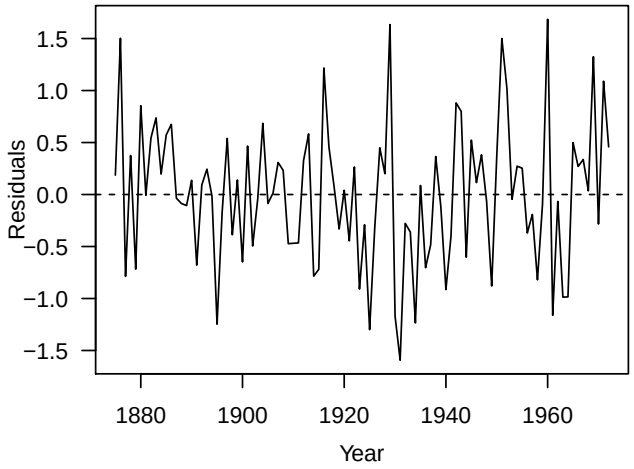
Coefficients:
         ar1      ar2  rep(1, length(LakeHuron))
      1.0048  -0.2913                    620.5115
s.e.  0.0976   0.1004                     15.5771
        years
      -0.0216
s.e.   0.0081

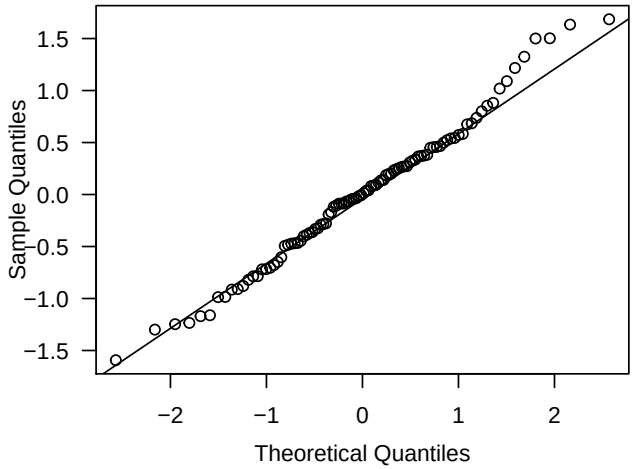
sigma^2 estimated as 0.4566:  log likelihood = -101.2,  aic = 212.4
```

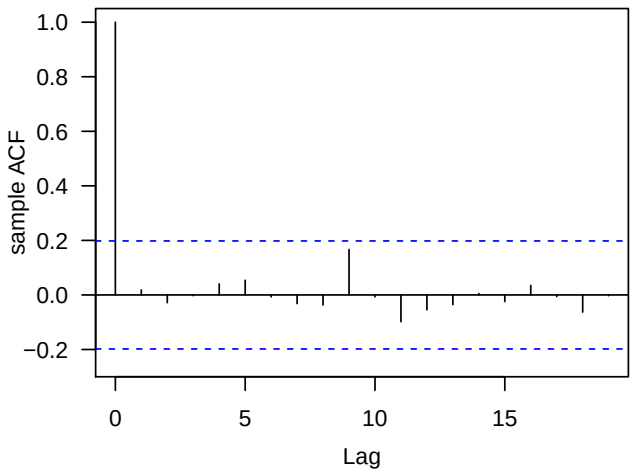
Focus: Estimated trend $\widehat{\beta}_1$, its uncertainty, and the fitted ARMA parameters.

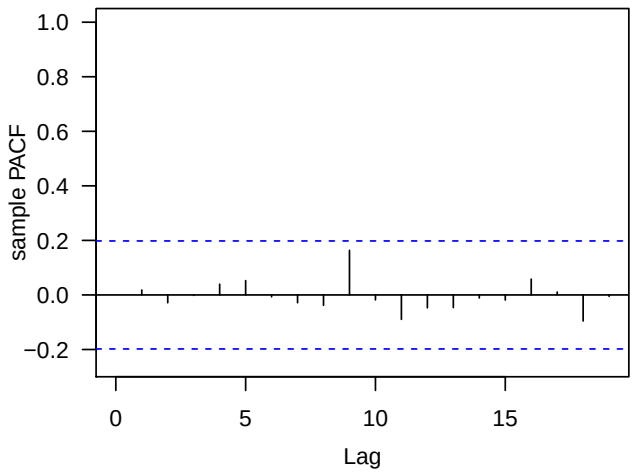
Lake Huron: Model Diagnostics

Are the model innovations $\approx$ white noise?










```
> plot.residuals(years, resid(mle), xlab = "Year", ylab = "Residuals")
```

```
        Box-Ljung test

data:  y
X-squared = 6.2088, df = 19, p-value = 0.9974
```

Lake Huron: Comparing Estimation Results

Regression slope β_1:

Method	Lower 95%	Estimate	Upper 95%
OLS	-0.0322	-0.0242	-0.0162
MLE	-0.0374	-0.0216	-0.0057

AR parameters:

Parameter	Method	Lower 95%	Estimate	Upper 95%
ϕ_1	Two-step	0.813	1.005	1.196
	MLE	0.813	1.005	1.196
ϕ_2	Two-step	-0.489	-0.293	-0.096
	MLE	-0.488	-0.291	-0.095

Key Findings

Both methods indicate a declining trend. Accounting for serial correlation gives a wider confidence interval for the slope.

Unit Root Tests: Stationary or Nonstationary?

Consider an AR(1) process:

$$X_t = \phi X_{t-1} + Z_t, \qquad Z_t \sim \mathrm{WN}(0, \sigma^2).$$

Condition	**Behavior**
$\lvert\phi\rvert < 1$	Stationary AR(1) process
$\phi = 1$	Random walk (unit root)

- A unit root test examines

$$\boxed{H_0 : \phi = 1 \quad \text{vs.} \quad H_a : \lvert\phi\rvert < 1.}$$

- Key question: Can we distinguish a stationary process from a random walk?

Dickey–Fuller Test: Reparameterization

Start with the AR(1) model with a constant:

$$X_t = c + \phi X_{t-1} + Z_t.$$

Subtract X_{t-1} from both sides:

$$\begin{aligned} \Delta X_t &= c + (\phi - 1) X_{t-1} + Z_t \\ &= c + \delta X_{t-1} + Z_t, \end{aligned}$$

where $\delta = \phi - 1$.

The Dickey–Fuller (DF) test considers

$$\boxed{H_0 : \delta = 0 \quad \text{vs.} \quad H_a : \delta < 0.}$$

- Exercise: Verify the reparameterization.
- Under H_0, the process has a unit root.

Dickey–Fuller Test: Test Statistic

Fit the regression by ordinary least squares:

$$\Delta X_t = c + \delta X_{t-1} + Z_t.$$

The DF test statistic is

$$\boxed{T_{\mathrm{DF}} = \frac{\hat{\delta}}{\widehat{\mathrm{SE}}(\hat{\delta})}.}$$

- Why not use the usual t-test?
  Under H_0, X_{t-1} is nonstationary, so the usual Student-t reference distribution is invalid.
- Instead, use the Dickey–Fuller distribution to obtain critical values or p-values.
- Reject H_0 for sufficiently negative values of T_{DF}.

Dickey–Fuller Test: Critical Values

For the DF regression **with an intercept**:

$$\Delta X_t = c + \delta X_{t-1} + Z_t.$$

Significance level α	0.01	0.05	0.10
Critical value C_α	-3.43	-2.86	-2.57

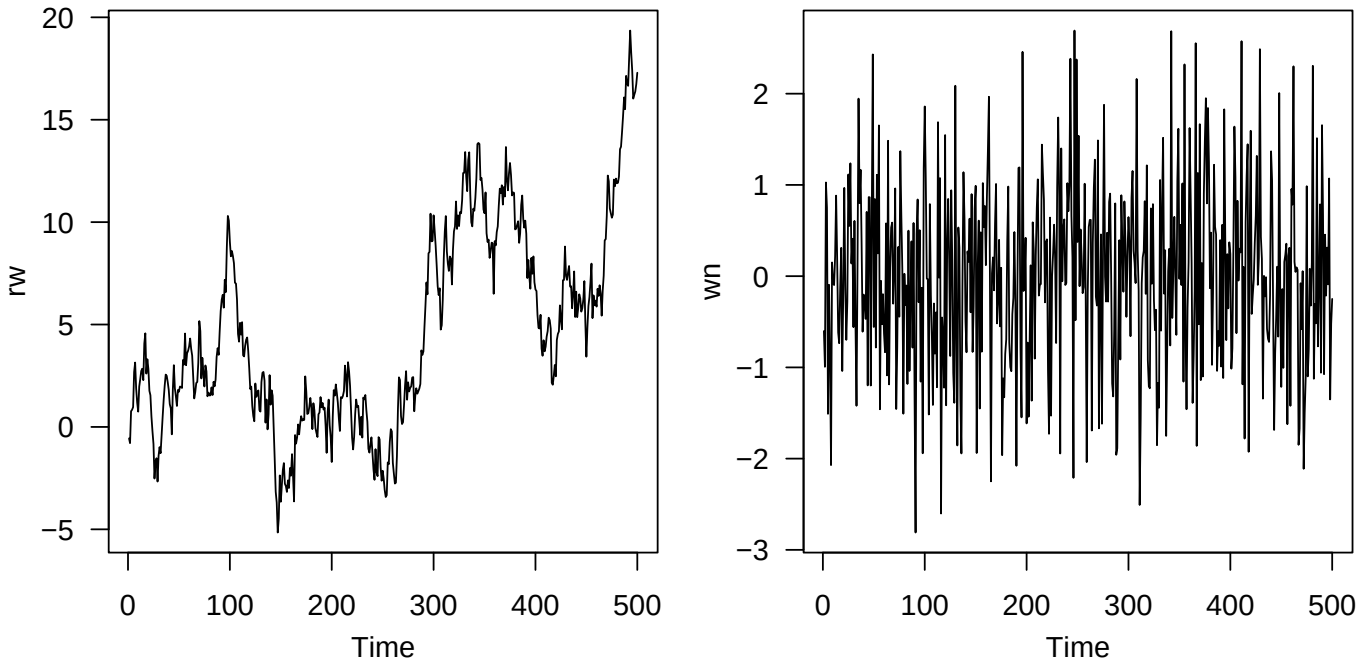
- Decision rule: Reject H_0 if $T_{\mathrm{DF}} < C_\alpha$.
- Critical values depend on whether the regression includes a constant and/or deterministic trend.
- These are approximate asymptotic critical values for the intercept-only specification.

Unit Root Tests: Simulated Examples

Compare two simulated processes:

$$\text{Random walk:} \quad X_t = X_{t-1} + Z_t, \quad \phi = 1,$$

$$\text{White noise:} \quad X_t = Z_t, \quad \phi = 0.$$



Question: Which series should provide stronger evidence against the unit-root null?

Dickey–Fuller Test: Random Walk Example

Simulated random walk: $\phi = 1$.

```
> diff.rw <- diff(rw); n <- length(rw)
> ys <- diff.rw; xs <- rw[1:(n-1)]
> ols.rw <- lm(ys ~ xs); summary(ols.rw)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.10125    0.05973   1.695   0.0906 .
xs          -0.01438    0.00899  -1.600   0.1102
```

Interpretation: Do we have sufficient evidence to reject the unit-root null?

Dickey–Fuller Test: White Noise Example

Simulated white noise: $\phi = 0$.

```
> diff.wn <- diff(wn)
> ys <- diff.wn; xs <- wn[1:(n-1)]
> ols.wn <- lm(ys ~ xs); summary(ols.wn)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.001138   0.045329  -0.025     0.98
xs          -1.002420   0.044843 -22.354   <2e-16
```

Interpretation: Does the test provide evidence of stationarity?

Augmented Dickey–Fuller (ADF) Test

The DF test assumes an AR(1) model with white-noise errors. For more general serial dependence, the ADF test adds lagged differences:

$$\Delta X_t = c + \delta X_{t-1} + \sum_{j=1}^{k} \gamma_j \Delta X_{t-j} + e_t.$$

- Test $H_0 : \delta = 0$ against $H_a : \delta < 0$.
- Lagged differences help account for serial correlation in the errors.
- The ADF statistic also uses nonstandard critical values.

Augmented Dickey–Fuller Test in R

Random Walk

```
> library(tseries)
> adf.test(rw)

        Augmented Dickey-Fuller Test

data:  rw
Dickey-Fuller = -1.9203, Lag order = 7, p-value =
0.612
alternative hypothesis: stationary
```

White Noise

```
> adf.test(wn)
Warning in adf.test(wn) : p-value smaller than printec

        Augmented Dickey-Fuller Test

data:  wn
Dickey-Fuller = -7.8953, Lag order = 7, p-value =
0.01
alternative hypothesis: stationary
```

- If $p < 0.05$, reject the unit-root null.
- Rejecting H_0 provides evidence in favor of stationarity
- Failing to reject H_0 does not prove the existence of a unit root.

From One Time Series to Two

So far: We have studied how a time series depends on its own past:

$$X_t \longleftarrow X_{t-1}, X_{t-2}, \ldots$$

From One Time Series to Two

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Now: How is one time series related to another?

$$Y_t \longleftarrow X_t, X_{t-1}, X_{t-2}, \ldots$$

For example, the effect of X_t on Y_t may occur with a delay:

$$\boxed{X_{t-d} \longrightarrow Y_t}$$

Cross-correlation helps identify potential lead–lag relationships between two series.

Lagged Regression and Cross-Correlation

Consider the lagged regression model:

$$Y_t = \beta_0 + \beta_1 X_{t-d} + \varepsilon_t,$$

where $\{X_t\}$ is iid with variance σ_X^2, and $\{\varepsilon_t\}$ is independent white noise.

The cross-covariance function is

$$\gamma_{XY}(h) = \mathbb{E}\left[(X_{t+h} - \mu_X)(Y_t - \mu_Y)\right].$$

The cross-correlation function (CCF) is

$$\boxed{\rho_{XY}(h) = \frac{\gamma_{XY}(h)}{\sqrt{\gamma_X(0)\gamma_Y(0)}}.}$$

Question: At which lag should the CCF be largest in magnitude?

Interpreting the CCF: Lead and Lag

Recall:

$$Y_t = \beta_0 + \beta_1 X_{t-d} + \varepsilon_t.$$

Under the iid and independence assumptions,

$$\gamma_{XY}(h) = \begin{cases} \beta_1 \sigma_X^2, & h = -d, \\ 0, & h \neq -d. \end{cases}$$

Thus,

$$\rho_{XY}(-d) = \frac{\beta_1 \sigma_X}{\sqrt{\beta_1^2 \sigma_X^2 + \sigma_\varepsilon^2}}.$$

$h = -d$ means that X leads Y by d time units.

Important: Always check the CCF lag convention before interpreting the sign of a lag.

Lagged Regression: A Simulated Example

Consider

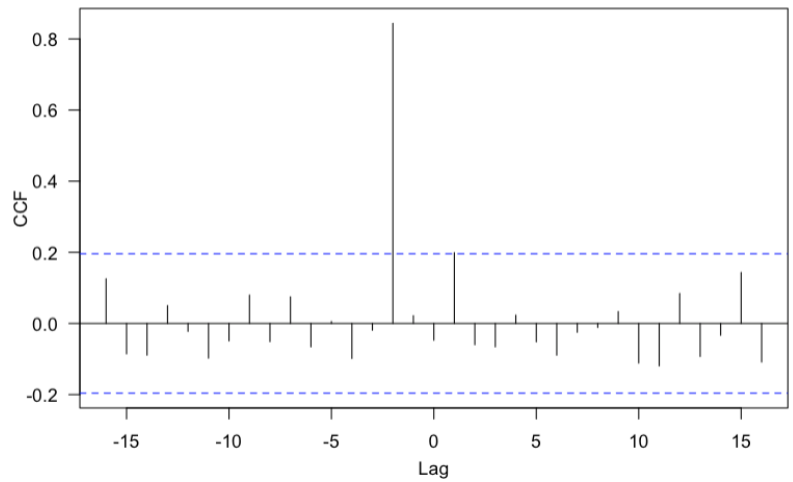
$$\boxed{Y_t = X_{t-2} + \varepsilon_t,}$$

where

$$X_t \overset{iid}{\sim} N(0,1), \qquad \varepsilon_t \overset{iid}{\sim} N(0, 0.25),$$

with the two sequences independent.
The theoretical CCF is

$$\rho_{XY}(h) = \begin{cases} 1/\sqrt{1.25} \approx 0.8944, & h = -2, \\ 0, & \text{otherwise.} \end{cases}$$

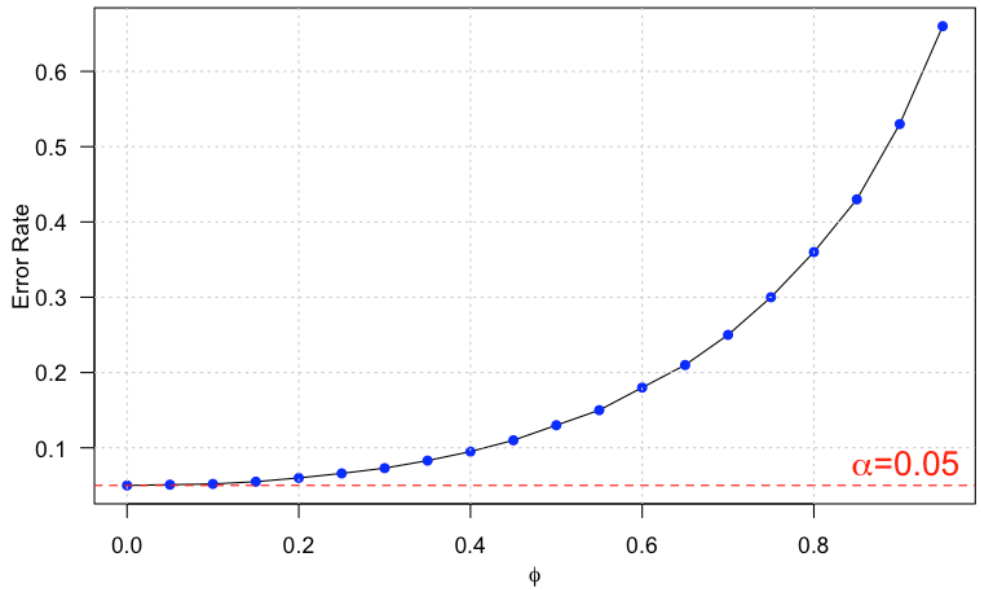


Why Raw Cross-Correlation Can Mislead

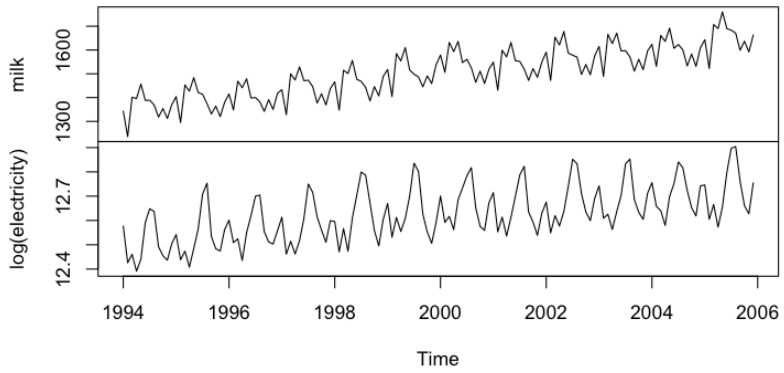
Two unrelated time series may exhibit large sample cross-correlations due to temporal dependence.

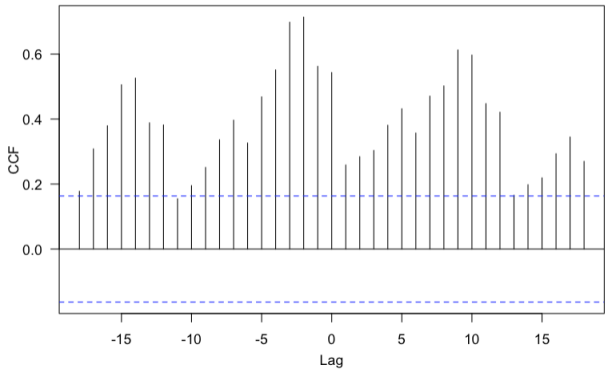
- Strong autocorrelation can produce apparently significant CCF values.
- The usual reference lines $\pm 1.96/\sqrt{n}$ may be unreliable.
- Shared trends or seasonality can also produce misleading correlations.

Example: Two independent AR(1) processes.



Spurious Correlation: Milk and Electricity





- **Observed Correlation**: Milk production and electricity usage show a high correlation due to shared seasonal patterns
- **Temporal Dependence**: Both series exhibit seasonality and autocorrelation, making raw correlations misleading
- **Key Takeaway**: Spurious correlations highlight the need for detrending and deseasonalizing in time series analysis

Prewhitening: Removing Serial Dependence

Goal: Reduce the internal serial dependence of X_t before examining its relationship with Y_t.
Suppose an ARMA model for X_t is

$$\phi(B)X_t = \theta(B)u_t,$$

where $\{u_t\}$ is approximately white noise.
Define the prewhitening filter:

$$\psi(B) = \frac{\phi(B)}{\theta(B)}.$$

Apply the **same filter** to both series:

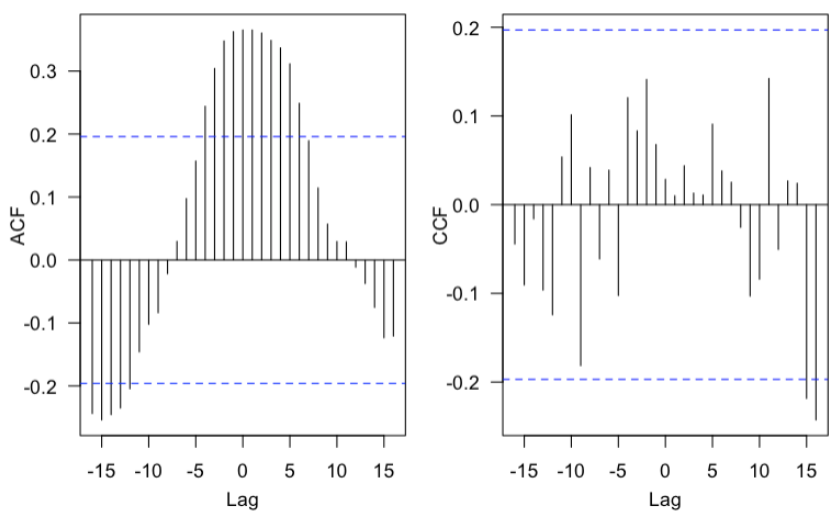
$$\boxed{\begin{aligned}\widetilde{X}_t &= \psi(B)X_t,\\ \widetilde{Y}_t &= \psi(B)Y_t.\end{aligned}}$$

Then examine $\mathrm{CCF}(\widetilde{X}_t, \widetilde{Y}_t)$.

Prewhitening: Step-by-Step Procedure

1. Fit an appropriate ARMA model to the input series X_t.
2. Filter X_t to obtain approximately white-noise innovations.
3. Apply the **same estimated filter** to the output series Y_t.
4. Compute the CCF between the two filtered series.

```
x <- arima.sim(n = 100, list(ar = 0.9))
y <- arima.sim(n = 100, list(ar = 0.9))
par(las = 1, mgp = c(2.2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6), mfrow = c(1, 2))
ccf(x, y)
prewhiten(x, y)
```



Prewhitening: Milk and Electricity Data

R implementation:

```
> me.dif = ts.intersect(diff(diff(milk, 12)),
+ diff(diff(log(electricity), 12)))
> prewhiten(as.vector(me.dif[, 1]), as.vector(me.dif[, 2]), ylab = 'CCF')
> par(las = 1, mgp = c(2.2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6))
> prewhiten(as.vector(me.dif[, 1]), as.vector(me.dif[, 2]), ylab = 'CCF')
```

CCF after prewhitening:

